

Joyita Banerjee Ganguly

Department of Mathematics, Central
Michigan University
Mt Pleasant, MI 48859
Phone: +1-517-243-4799
Email: baner1j@cmich.edu

Education

Doctor of Philosophy

2023–Present (Expected May 2027)

Central Michigan University, Mt Pleasant, Michigan, USA

Research Area: Differential Geometry, Complex Analysis

Advisor: Dr. Debraj Chakrabarti

Dissertation Title: Volume of domains bounded by Hermitian quadrics in complex projective space

Master of Arts in Mathematics

2025–2026

Central Michigan University, Mt Pleasant, Michigan, USA

UPE Project II

2016–2017

University of Hyderabad, Hyderabad, India

Research Area: Differential Equations

Project Title: Oscillation Theory of Dynamic Equations on Time Scale,

Supervisor: Dr. S. Panigrahi

Master of Science in Applied Mathematics

2014–2016

University of Hyderabad, Hyderabad, India

Bachelor of Science in Mathematics (First Class Honours)

2011–2014

University of Calcutta, Kolkata, India

Professional and Teaching Experience

- **Graduate Teaching Assistant** (2023 – Present), Department of Mathematics, Central Michigan University, MI. Served as Instructor of Record for all courses taught.

Course	Year(s)
MTH 107 College Algebra, in person	Fall 2026 (2 sections)
MTH 334 Differential Equations (Teaching Internship)	Spring 2026
MTH 101 Beginning Algebra	Spring 2024, Spring 2025
MTH 107 College Algebra (online asynchronous)	Spring 2024
MTH 105 Intermediate Algebra	Fall 2023, Fall 2024 (2 sections)
MTH 223 Linear Algebra and Matrix Theory (Teaching Internship, scheduled)	Spring 2027

- **Math Tutor at Central Michigan University:** Tutored undergraduate students at the Mathematics Assistance Center (MAC) every semester; worked full-time at the MAC in Fall 2025. The MAC is a free math help center for undergraduate students.
- **Teaching Experience in India** (2017 – 2023): Taught senior high school and IIT/Mathematics Olympiad aspirant students Mathematics, conducted assessments and mock examinations, and prepared termly and status reports, across the following institutes – Aakash Educational Services Limited (Assistant Lecturer, Sep. 2021 – Mar. 2023), Vedantu (Subject Matter Expert and Team Lead, Oct. 2019 – Aug. 2021; led a team of 25 teachers and content writers as reporting manager), Motion Education (Senior Faculty – IIT Division, Mar. 2019 – Oct. 2019), and Sri Chaitanya Junior College (Faculty – IIT and Mathematics Olympiad Division, Jul. 2017 – Feb. 2019).

Student Mentoring

- **Volunteer Mentor, MathaPalooza, Department of Mathematics, Central Michigan University:** Mentored undergraduate students at this semi-annual math help session held each semester on the Sunday before final exam week.

Service

Incoming Graduate Teaching Assistant Mini Teaching Session: *Observer*.

Fall 2026

Department of Mathematics, Central Michigan University

American Mathematical Society Graduate Student Chapter: *Treasurer*.

Fall 2025– Spring 2026

Department of Mathematics, Central Michigan University

American Mathematical Society Graduate Student Chapter: *Secretary*.

Fall 2024– Spring 2025

Department of Mathematics, Central Michigan University

Professional Development

Graduate Teaching Assistant Workshop

Fall 2023

This is a week-long face-to-face training for the new graduate teaching assistants, and it concludes with a mini teaching session.

The Office of Curriculum and Instructional Support GTA Series

Fall 2026

These face-to-face sessions are designed to enhance teaching transparency, creativity, and engagement, focusing on achieving instructional excellence through conversations about modern classroom practices under the theme “Building Excellence.”

Incorporating AI into Assignments

Fall 2026

An interactive hybrid workshop on designing assignments that intentionally incorporate generative AI tools in ways that support learning, critical thinking, and disciplinary skill development, with guided time to revise an existing assignment.

Publications

- **Banerjee Ganguly, J.,** Chakrabarti, D., Mainkar, M. *Volume of the domain bounded by a Hermitian quadric in complex projective space*. Preprint, arXiv:2601.07077 [math.MG], January 2026. <https://doi.org/10.48550/arXiv.2601.07077>

Awards and Honors

- **American Mathematical Society Graduate Student Travel Grant**, Fall 2026
- **Graduate Research Assistantship**, Central Michigan University, Summer 2026.
- **Graduate Research Assistantship**, Central Michigan University, December 2025.
- **Grand Integrator – Winner at the Annual Integration Bee Competition**, Central Michigan University, November 2023.
- **Graduate Research Assistantship**, Central Michigan University, Fall 2023.
- **Awarded NPTEL Certificate, Introduction to CFD**, IIT Madras, India (Aug. 2022 – Jan. 2023).
- **Qualified IIT Joint Admission Test for Masters (JAM)**, 2014.

Presentations and Talks Given

- *Volume Computation of Hermitian Quadric in Complex Projective Space*, AMS Sectional Meeting, George Washington University, DC, USA October 3, 2026.
- *Volume Computation of Hermitian Quadric in Complex Projective Space*, Invited talk, Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI), Belur, India, July 22, 2026.
- *Volume of the Domain Bounded by a Hermitian Quadric in Complex Projective Space*, poster presentation, SCREE (Student Creative and Research Endeavors Exhibition), Central Michigan University, April 15, 2026.
- *The Geometry of Complex Projective Space*, Graduate Student Seminar, Central Michigan University, March 17, 2026.
- *Symmetric Polynomials*, Graduate Student Seminar, Central Michigan University, February 17, 2026.

Conferences and Workshops Attended

- **Joint Mathematics Meetings (JMM) 2027**, Chicago, IL (scheduled), January 2027.
- **AMS Eastern Fall Sectional Meeting**, George Washington University, Washington, DC (scheduled), October 3–4, 2026.
- **MAA Webinar: AI in the Classroom: What’s Working (and What’s Not)**, Virtual, May 20–21, 2026.
- **MAA Michigan Sectional Meeting**, Saginaw Valley State University, March 27–28, 2026.
- **Indian Women and Mathematics Annual Conference**, University of Hyderabad, India, Jun. 2016.
- **International Conference for Analytic & Algebraic Geometry**, University of Hyderabad, India, Mar. 2015.
- **National Seminar (UGC): Dynamical System – Its Application and Consequences**, St. Paul’s Cathedral Mission College, Kolkata, India, Sep. 2013.

Skills

- **Computer Programming:** MATLAB, Python, Wolfram Mathematica, C, C++, L^AT_EX, Linux
- **Others:** MS Office (Word, Excel, PowerPoint)

Professional Membership

- Member, American Mathematical Society (AMS).
- Member, Mathematical Association of America (MAA).

Languages

English, Hindi, and Bengali(MT): Proficient in Listening, Speaking, Reading, and Writing

Additional Information

- **Website:** <https://joyitabanerjeeanguly.github.io/index.html>
- **Orcid:** <https://orcid.org/0009-0001-2803-884X>
- **Researchgate:** <https://www.researchgate.net/profile/Joyita-Banerjee-Ganguly>
- **LinkedIn:** <https://www.linkedin.com/in/joyita-banerjee-ganguly-268b65179>